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United States Patent Application Publication

20250271192

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

Liu; Zhicheng et al.

ICE MAKING MACHINE

Abstract

An ice-making machine includes an ice-making tray assembly that has multiple ice trays. An ice-turning motor is fixedly connected to one end of the ice-making tray assembly. The ice-turning motor is used to drive the ice-making tray assembly to turn. The ice-making machine also includes an air duct that has an air inlet and an air outlet. The air outlet has an inclined baffle arranged at an angle along the cold air flow direction. The inclined baffle gradually approaches the ice-making tray assembly at an acute angle relative to a horizontal surface of the icemaking tray assembly.

Inventors: Liu; Zhicheng (Anhui, CN), Liu; Dian (Hefei City, CN), Li; Jiaqi (Anhui, CN), Rui; Qunna (Hefei City, CN), Liu; Yuxiang (Hefei, CN)

Applicant: WHIRLPOOL CORPORATION (BENTON HARBOR, MI)

Family ID: 1000008475340

Assignee: WHIRLPOOL CORPORATION (BENTON HARBOR, MI)

Appl. No.: 19/059885

Filed: February 21, 2025

Foreign Application Priority Data

CN

202420335308.0

Feb. 22, 2024

Publication Classification

Int. Cl.: F25C1/10 (20060101)

U.S. Cl.:

CPC F25C1/10 (20130101); F25C2400/10 (20130101); F25C2700/12 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to Chinese Utility Model Patent Application No. CN 202420335308.0, filed on Feb. 22, 2024, published as CN 222230984 U on Dec. 24, 2024, and entitled “An ice making machine and a refrigerator,” the entire disclosure of which is hereby incorporated by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to the technical field of refrigerators, and more specifically relates to an ice making machine and a refrigerator.

BACKGROUND

[0003] With the continuous improvement of living standards, refrigerators have become an important appliance. Refrigerators use refrigeration to maintain an interior cabinet at a low temperature, which is beneficial to preserve food. Refrigerators can also make ice by setting up a simple ice-making device, which is extremely convenient and can meet a users' demand for ice cubes. However, existing refrigerator ice makers still have several defects. For example, when assembling the ice making machine used in traditional refrigerators, the assembly method is complicated, which makes it difficult to assemble. Also, in traditional refrigerators, air inlets of the ice machine may be designed to be flush with the surface of the ice tray, which allows cold air to flow horizontally across the surface of the ice tray. Since the level of the ice-making surface is lower than the surface of the ice tray, and the cold air flows horizontally, eddy currents will appear in the ice tray close to the air outlet, which results in insufficient heat exchange and a longer freezing time for the ice cubes.

[0004] Compared with existing technology, the present disclosure realizes the rapid assembly of the frame and the box of the ice machine through simple assembly components. Through the setting of the inclined baffle at the air outlet of the induced air duct, it solves the problem of long freezing time of ice caused by insufficient heat exchange problem, thus shortening the ice making cycle.

SUMMARY

[0005] According to an aspect of the present disclosure, an ice making machine includes an ice making machine frame that is provided with an ice-making tray assembly that has multiple ice trays. An ice-turning motor is fixedly connected to one end of the ice tray and is used to drive the ice tray to turn over. An induced air duct includes an air inlet and an air outlet. The air outlet is provided with a sloped baffle. The sloped baffle is arranged at an angle and is perpendicular to a surface of the ice tray along a cold air flow direction, in the direction gradually closer to the ice tray.

[0006] According to another aspect of the present disclosure, an ice-making machine includes an ice-making tray assembly that has multiple ice trays. An ice-turning motor is fixedly connected to one end of the ice-making tray assembly. The ice-turning motor is used to drive the ice-making tray assembly to turn. The ice-making machine also includes an induced air duct that an air inlet and an air outlet and an ice-making machine frame that has an ice detection rod. The air outlet includes an inclined baffle arranged at an angle along the cold air flow direction. The inclined baffle gradually approaches the ice-making tray assembly in a direction perpendicular to a horizontal surface of the ice-making tray assembly. The ice detection rod is installed on the ice-turning motor and the ice-making machine frame is installed in a box of a freezing chamber of a refrigerator. Assembly components are distributed on a contact surface between the ice-maker frame and the box to fix the ice-maker frame to the box. The assembly components include a clamping slot, a fixed hanging column, and an elastic limiter.

[0007] According to another aspect of the present disclosure, an ice-making machine includes an

ice-making tray assembly that has multiple ice trays. An ice-turning motor is fixedly connected to one end of the ice-making tray assembly. The ice-turning motor is used to drive the ice-making tray assembly to turn. The ice-making machine also includes an air duct that has an air inlet and an air outlet. The air outlet has an inclined baffle arranged at an angle along the cold air flow direction. The inclined baffle gradually approaches the ice-making tray assembly at an acute angle relative to a horizontal surface of the ice making tray assembly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] In the drawings:

[0009] FIG. 1 is a schematic diagram of the overall structure of a refrigerator appliance of the present disclosure;

[0010] FIG. 2 is a schematic structural diagram of the refrigerator of the present disclosure;

[0011] FIG. 3 is a schematic structural diagram of the ice machine frame of the present disclosure;

[0012] FIG. 4 is a schematic assembly diagram of the frame and box of the ice making machine of the present disclosure; and

[0013] FIG. 5 is a partial assembly diagram of the frame and box of the ice making machine of the present disclosure.

[0014] The components in the figures are not necessarily to scale, emphasis instead being placed upon illustrating the principles described herein.

DETAILED DESCRIPTION

[0015] The present illustrated embodiments reside primarily in combinations of method steps and apparatus components related to an ice making machine of a refrigeration appliance. Accordingly, the apparatus components and method steps have been represented, where appropriate, by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein. Further, like numerals in the description and drawings represent like elements.

[0016] For purposes of description herein, the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” and derivatives thereof shall relate to the disclosure as oriented in FIG. 1. Unless stated otherwise, the term “front” shall refer to the surface of the element closer to an intended viewer, and the term “rear” shall refer to the surface of the element further from the intended viewer. However, it is to be understood that the disclosure may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

[0017] The terms “including,” “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises a . . .” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element

[0018] Referring to FIG. 1, a refrigeration appliance with a freezing chamber **10** is shown. The freezing chamber **10** includes a box **11** and an ice making machine **111**. An ice making machine frame **20** (FIG. 2) is operably coupled with the box **11** of the freezing chamber **10**.

[0019] As shown in FIGS. 2 and 3, the ice making machine **111** is illustrated The ice making

machine **111** includes an ice making machine frame **20**. The ice making machine frame **20** is provided with an ice-making tray assembly **21** (FIG. **4**), an ice turning motor **22**, and a detector. [0020] The ice-making tray assembly **21** has a plurality of ice trays **211**. One end of the ice-making tray assembly **21** is operably coupled with the ice machine frame **20** and the other end is fixedly connected to the ice turning motor **22**. The ice-making tray assembly **21** may be fixedly coupled with the ice making machine frame **20** and the other end may be fixedly coupled with the ice-turning motor **22**.

[0021] The ice-turning motor **22** is installed on the ice-making machine frame **20** together with the ice-making tray assembly **21**. The ice-turning motor **22** is fixed through two holes by elastic features on the ice-making machine frame **20**. The ice turning motor **22** may be energized to drive the ice-making tray assembly **21** to rotate or swing, causing ice cubes in the ice tray **211** to fall. An ice detecting rod **23** is installed on the ice turning motor **22**. The ice detecting rod **23** is configured to detect the formation of ice cubes.

[0022] With reference now to FIG. **4**, an air duct **24** has an air inlet **241** located on a side of the ice-making tray assembly **21** and an air outlet **242**. Cold air flows from bottom to top in a vertical direction of the ice making machine frame **20**. The air outlet **242** is provided with an inclined baffle **2421**. The inclined baffle **2421** is arranged in an angled or inclined manner and gradually approaches the ice-making tray assembly **21** along the direction of cold air flow. The inclined baffle **2421** is used to prevent at least a portion of the cold air from flowing along the surface of the ice tray **211**. While flowing horizontally, another portion of the cold air is guided to flow obliquely toward the ice tray **211** close to the induced draft duct **24**. The inclined baffle **2421** is provided to solve the problem of eddy currents that can develop close to the air outlet **242** in traditional ice machines, which in turn causes insufficient heat exchange and a long freezing time for the ice cubes. The inclined baffle **2421** is inclined from top to bottom along the cold air flow direction and may approach a horizontal surface of the ice-making tray assembly **21** at an acute angle. The angle range with the horizontal surface is between 20° and 60°. In some instances, the inclined baffle **2421** may gradually approach the horizontal surface of the ice-making tray assembly **21** at a perpendicular angle relative to the horizontal surface. A distance between the lowest point of the inclined baffle **2421** and the horizontal surface of the ice-making tray assembly **21** ranges from 3 cm to 9 cm.

[0023] Referring now to FIG. **5**, assembly components **25** are distributed on the contact surface between the ice maker frame **20** and a body of the box **11** to fix the ice maker frame **20** and the box **11**. As shown in FIGS. **4** and **5**, the assembly components **25** include a slot **251**, a fixed hanging post **252**, and an elastic stopper **253**. In some instances, the slot **251** has a U-shaped opening. Further, the slots **251** may be evenly distributed at the edge of the contact surface between the ice machine frame **20** and the box **11**. The contact surface between the box **11** and the ice machine frame **20** is provided with fixing holes for engaging the slots **251**. The fixed hanging post **252** is pushed horizontally into the slot **251** along the inward direction of the opening of the slot **251** to fix the ice maker frame **20** and the box **11**. The fixed hanging post **252** has a T-shaped structure, the outer surface of the extending end in the vertical direction has a concave-convex structure, and the concave portion of the fixed hanging post **252** engages with the slot **251**. The elastic stopper **253** is disposed at the opening of the slot **251** and is tilted into the slot **251** to resist the fixed hanging post **252**. The configuration of the assembly components **25** has a simple structure and reduces the installation process of the ice maker and the refrigerator box **11**.

[0024] According to an aspect of the present disclosure, an ice making machine includes an ice making machine frame that is provided with an ice tray that has multiple ice trays. An ice-turning motor is fixedly connected to one end of the ice tray and is used to drive the ice tray to turn over. An induced air duct includes an air inlet and an air outlet. The air outlet is provided with a sloped baffle. The sloped baffle is arranged at an angle and is perpendicular to a surface of the ice tray along a cold air flow direction, in the direction gradually closer to the ice tray.

[0025] According to still another aspect of the present disclosure, an angle range between an inclined baffle and a horizontal plane is between 20°-60°.

[0026] According to yet another aspect of the present disclosure, a distance between a lowest point of an inclined baffle and a horizontal surface of an ice tray ranges from 3 cm to 9 cm.

[0027] According to still another aspect of the present disclosure, an ice detection rod is also provided on an ice making machine frame. The ice detection rod is installed on an ice turning motor.

[0028] According to another aspect of the present disclosure, a refrigerator includes a freezing chamber and an ice making machine. An ice making machine frame of the ice making machine is installed on a box body of the freezing chamber.

[0029] According to another aspect of the present disclosure, assembly components are distributed on a contact surface between an ice maker frame and a box body to fix the ice maker frame and the box body.

[0030] According to another aspect of the present disclosure, an assembly includes a slot, a fixed hanging column, and an elastic stopper. The slot is evenly distributed at an edge of a contact surface between an ice machine frame and a box. The contact surface with the frame of the ice maker is provided with a fixing hole for fixing the slot by pushing the fixed hanging post horizontally into the slot along the inward direction of the slot opening to fix the ice maker, Ice machine frame, and box.

[0031] According to still another aspect of the present disclosure, a fixed hanging column has a T-shaped structure and an outer surface of an extending end in a vertical direction has a concave-convex structure. The concave portion of the fixed hanging column engages with the slot.

[0032] According to still another aspect of the present disclosure, an elastic stopper is disposed at an opening of the slot and tilts into the slot to resist the fixed hanging post.

[0033] According to another aspect of the present disclosure, an ice-making machine includes an ice-making tray assembly that has multiple ice trays. An ice-turning motor is fixedly connected to one end of the ice-making tray assembly. The ice-turning motor is used to drive the ice-making tray assembly to turn. The ice-making machine also includes an induced air duct that an air inlet and an air outlet and an ice-making machine frame that has an ice detection rod. The air outlet includes an inclined baffle arranged at an angle along the cold air flow direction. The inclined baffle gradually approaches the ice-making tray assembly in a direction perpendicular to a horizontal surface of the ice-making tray assembly. The ice detection rod is installed on the ice-turning motor and the ice-making machine frame is installed in a box of a freezing chamber of a refrigerator. Assembly components are distributed on a contact surface between the ice-maker frame and the box to fix the ice-maker frame to the box. The assembly components include a clamping slot, a fixed hanging column, and an elastic limiter.

[0034] According to another aspect of the present disclosure, an angle between an inclined baffle and a horizontal surface ranges from 20° to 60°.

[0035] According to another still aspect of the present disclosure, a distance between a lowest point of an inclined baffle and a horizontal surface of an ice-making tray assembly ranges from 3 cm to 9 cm.

[0036] According to yet another aspect of the present disclosure, a clamping slot is evenly distributed at an edge of a contact surface between an ice-making machine frame and a box.

[0037] According to another aspect of the present disclosure, a fixing slot is disposed at the contact surface between the box and the ice-making machine frame for fixing the clamping slot.

[0038] According to another aspect of the present disclosure, a fixed hanging column is configured to be pushed horizontally into a clamping slot along an inward direction of an opening of the clamping slot to fix the clamping slot, an ice-making machine frame, and a box.

[0039] According to still another aspect of the present disclosure, a fixed hanging column has a T shaped structure. An outer surface of an extending end in a vertical direction of the T-shaped

structure has a concave part and a convex part. The concave part of the T-shaped structure is engaged with a clamping slot.

[0040] According to yet another aspect of the present disclosure, an elastic limiter is disposed at an opening of a clamping slot and is inclined toward an inside of the clamping slot to resist the opening of the clamping slot.

[0041] According to another aspect of the present disclosure, an ice-making machine includes an ice-making tray assembly that has multiple ice trays. An ice-turning motor is fixedly connected to one end of the ice-making tray assembly. The ice-turning motor is used to drive the ice-making tray assembly to turn. The ice-making machine also includes an air duct that has an air inlet and an air outlet. The air outlet has an inclined baffle arranged at an angle along the cold air flow direction. The inclined baffle gradually approaches the ice-making tray assembly at an acute angle relative to a horizontal surface of the ice making tray assembly.

[0042] According to still another aspect of the present disclosure, an ice-making machine includes an ice-making machine frame that has an ice detection rod. The ice detection rod is installed on an ice-turning motor.

[0043] According to yet another aspect of the present disclosure, an ice-making machine includes assembly components that include a clamping slot, a fixed hanging column, and an elastic limiter.

[0044] According to another aspect of the present disclosure, an ice-making machine frame is installed in a box of a freezing chamber. A fixing slot is disposed at a contact surface between the box and the ice-making machine frame for fixing a clamping slot. It will be understood by one having ordinary skill in the art that construction of the described disclosure and other components is not limited to any specific material. Other exemplary embodiments of the disclosure disclosed herein may be formed from a wide variety of materials, unless described otherwise herein.

[0045] For purposes of this disclosure, the term “coupled” (in all of its forms, couple, coupling, coupled, etc.) generally means the joining of two components (electrical or mechanical) directly or indirectly to one another. Such joining may be stationary in nature or movable in nature. Such joining may be achieved with the two components (electrical or mechanical) and any additional intermediate members being integrally formed as a single unitary body with one another or with the two components. Such joining may be permanent in nature or may be removable or releasable in nature unless otherwise stated.

[0046] It is also important to note that the construction and arrangement of the elements of the disclosure as shown in the exemplary embodiments is illustrative only. Although only a few embodiments of the present innovations have been described in detail in this disclosure, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts or elements shown as multiple parts may be integrally formed, the operation of the interfaces may be reversed or otherwise varied, the length or width of the structures and/or members or connector or other elements of the system may be varied, the nature or number of adjustment positions provided between the elements may be varied. It should be noted that the elements and/or assemblies of the system may be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures, and combinations. Accordingly, all such modifications are intended to be included within the scope of the present innovations. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the desired and other exemplary embodiments without departing from the spirit of the present innovations.

[0047] It will be understood that any described processes or steps within described processes may be combined with other disclosed processes or steps to form structures within the scope of the

present disclosure. The exemplary structures and processes disclosed herein are for illustrative purposes and are not to be construed as limiting.

Claims

1-9. (canceled)

10. An ice-making machine, comprising: an ice-making tray assembly having multiple ice trays; an ice-turning motor fixedly connected to one end of the ice-making tray assembly, wherein the ice-turning motor is used to drive the ice-making tray assembly to turn; an induced air duct having an air inlet and an air outlet, the air outlet having an inclined baffle arranged at an angle along the cold air flow direction, wherein the inclined baffle gradually approaches the ice-making tray assembly in a direction perpendicular to a horizontal surface of the ice-making tray assembly; an ice-making machine frame having an ice detection rod, wherein the ice detection rod is installed on the ice-turning motor, wherein the ice-making machine frame is installed in a box of a freezing chamber of a refrigerator; and assembly components distributed on a contact surface between the ice-maker frame and the box to fix the ice-maker frame to the box, wherein the assembly components include a clamping slot, a fixed hanging column, and an elastic limiter.

11. The ice-making machine of claim 10, wherein the angle between the inclined baffle and the horizontal surface ranges from 20° to 60°.

12. The ice-making machine of claim 10, wherein a distance between a lowest point of the inclined baffle and the horizontal surface of the ice-making tray assembly ranges from 3 cm to 9 cm.

13. The ice-making machine of claim 10, wherein the clamping slot is evenly distributed at an edge of the contact surface between the ice-making machine frame and the box.

14. The ice-making machine of claim 10, wherein a fixing slot is disposed at the contact surface between the box and the ice-making machine frame for fixing the clamping slot.

15. The ice-making machine of claim 14, wherein the fixed hanging column is configured to be pushed horizontally into the clamping slot along an inward direction of an opening of the clamping slot to fix the clamping slot, the ice-making machine frame, and the box.

16. The ice-making machine of claim 10, wherein the fixed hanging column has a T-shaped structure.

17. The ice-making machine of claim 16, wherein an outer surface of an extending end in a vertical direction of the T-shaped structure has a concave part and a convex part.

18. The ice-making machine of claim 17, wherein the concave part of the T-shaped structure is engaged with the clamping slot.

19. The ice-making machine of claim 15, wherein the elastic limiter is disposed at the opening of the clamping slot and is inclined toward an inside of the clamping slot to resist the opening of the clamping slot.

20. An ice-making machine, comprising: an ice-making tray assembly having multiple ice trays; an ice-turning motor fixedly connected to one end of the ice-making tray assembly, wherein the ice-turning motor is used to drive the ice-making tray assembly to turn; and an air duct having an air inlet and an air outlet, the air outlet having an inclined baffle arranged at an angle along the cold air flow direction, wherein the inclined baffle gradually approaches the ice-making tray assembly at an acute angle relative to a horizontal surface of the ice-making tray assembly.

21. The ice-making machine of claim 20, wherein the angle between the inclined baffle and the horizontal surface ranges from 20° to 60°.

22. The ice-making machine of claim 20, further comprising: an ice-making machine frame having an ice detection rod, wherein the ice detection rod is installed on the ice-turning motor.

23. The ice-making machine of claim 22, further comprising: assembly components including a clamping slot, a fixed hanging column, and an elastic limiter.

24. The ice-making machine of claim 23, wherein the ice-making machine frame is installed in a

box of a freezing chamber, and wherein a fixing slot is disposed at a contact surface between the box and the ice-making machine frame for fixing the clamping slot.

25. The ice-making machine of claim 23, wherein the fixed hanging column has a T-shaped structure.

26. The ice-making machine of claim 25, wherein an outer surface of an extending end in a vertical direction of the T-shaped structure has a concave part and a convex part.

27. The ice-making machine of claim 26, wherein the concave part of the T-shaped structure is engaged with the clamping slot.

28. The ice-making machine of claim 23, wherein the elastic limiter is disposed at an opening of the clamping slot and is inclined toward an inside of the clamping slot to resist the opening of the clamping slot.
